$+$ $\times$

$-$ $\div$

RECAP

WE CONSIDERED SINGLE-PHOTON INTERFERENCE IN A TWO-BEAM SYSTEM

$\longrightarrow$ "UPPER"

$\longrightarrow$ "LOWER"

2 BEAMS $\Rightarrow$ 2 ALTERNATIVES

$v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ VECTOR OF PROB. AMPLITUDE $\quad P_{UP} = |\alpha|^2$, $P_{LOW} = |\beta|^2 \longrightarrow v^\dagger v = 1$

NORMALIZATION CONDITION: $|\alpha|^2 + |\beta|^2 = 1$

"DAGGER" PUGNALE

$v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad v^\dagger = (\alpha^* \quad \beta^*) \leftarrow$ HERMITIAN CONJUGATE

=

(TRANSPOSE + CONJUGATE)

↓

I TRANSPOSE THE VECTOR

$v^\dagger v = (\alpha^* \quad \beta^*) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = |\alpha|^2 + |\beta|^2$

ANY PHYSICALLY POSSIBLE LINEAR OPTICAL ELEMENT IN A TWO-BEAM INTERFEROMETER IS REPRESENTED BY A 2x2 UNITARY MATRIX

$R^\dagger R = R R^\dagger = 1$ WHERE $R^\dagger = (R^T)^*$

IF $v^\dagger v = 1$ AND $v' = Rv \Rightarrow (v')^\dagger v' = 1$

IMP: A MATRIX M THAT IS THE PRODUCT OF UNITARY MATRICES IS UNITARY

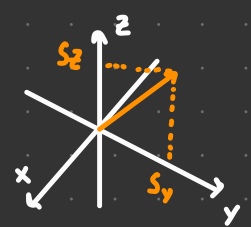
INFORMATION AND PHYSICAL SYSTEMS

IT IS A PROPERTY OF THE PARTICLE ITSELF

• SPIN 1/2 $\quad$ A PARTICLE CAN HAVE AN INTRINSIC ANGULAR MOMENTUM CALLED SPIN

E.G. ELECTRONS DO HAVE A SPIN, BUT THEY APPEAR TO BE ENTIRELY POINT-LIKE, WITHOUT ANY SPATIAL EXTENT AT ALL

ELECTRONS, PROTONS AND NEUTRONS $\rightarrow$ ALL EXAMPLE OF SPIN 1/2 PARTICLES



SUPPOSE WE MEASURE THE z-COMPONENT S_z OF THE SPIN ANGULAR MOMENTUM FOR ONE OF THESE PARTICLES

z NOT SPECIAL, SAME RESULT WOULD OCCOUR IN ANY OTHER DIRECTION

$\hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34}$ Js $\quad$ UNIT OF ANGULAR MOMENTUM

↳ IT TURNS OUT THAT RESULT IS EITHER $+\hbar/2$ OR $-\hbar/2$, NO MATTER THE SPIN ORIENTATION

CAN WE PROVE IT TRUE? HOW CAN A COMPONENT OF A PARTICLE'S INTRINSIC ANGULAR MOMENTUM BE MEASURED?

OTTO STERN AND WALTHER GERLACH

SPIN HARD TO PROBE DIRECTLY, ONE CAN SHOW THAT THERE'S A MAGNETIC MOMENT ASSOCIATED W/ A SPIN

$\vec{\mu} = \gamma \vec{s}$ WHERE γ IS CALLED GYROMAGNETIC RATIO

• MEASURING MAGNETIC MOMENTUM MUCH EASIER, FOR ONE CAN SHOW THAT ASSOCIATED ENERGY IN A MAGNETIC FIELD B IS

$$E = -\vec{\mu} \cdot \vec{B}$$

GIVEN A MAGNETIC FIELD IN z-DIRECTION

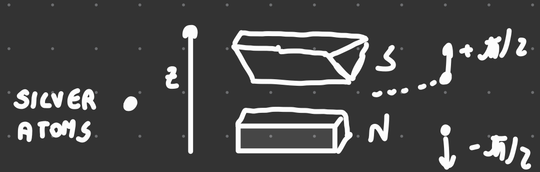
$$E = \mu_z B_z$$

• STERN AND GERLACH REALIZED THAT TRAJECTORY OF AN ATOM IS A NON-UNIFORM MAGNETIC FIELD DEPEND ON THIS ENERGY

$$\vec{F} = \mu_z \frac{dB_z}{dz} \hat{z}$$

MEASURING SPIN OF A PARTICLE

THEY CONSTRUCTED AN APPARATUS TO CREATE A NON-UNIFOR MAGNETIC FIELD ALONG A GIVEN DIRECTION (z)



TWO POSSIBLE COMPONENTS OF MAGNETIC FIELD + OR −

RESULT PROVED THAT SPIN IS QUANTIZED
↳ ONLY TWO POSSIBLE VALUES (IN THIS CASE)

PHOTONS

$\longrightarrow$ UPPER

$\longrightarrow$ LOWER

SPIN 1/2 PARTICLE

$\uparrow\downarrow$

2 SPIN ORIENTATION

$\downarrow$

2 ALTERNATIVES

α (SPIN UP) + β (SPIN DOWN)

$\updownarrow$

$\alpha z_+ + \beta z_-$

$z_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ UPPER BEAM

$z_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ LOWER BEAM

$z_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ COMPONENT OF THIS VECTOR DO NOT HAVE GEOMETRIC MEANING

ANGULAR MOMENTUM IS A THREE-DIMENSIONAL VECTOR

THERE CAN BE A SUPERPOSITION OF z_+ AND z_- WHICH CORRESPOND TO THE SITUATION IN WHICH THE PARTICLE HAS A WELL DEFINED SPIN COMPONENT IN A PARTICULAR DIRECTION

E.g. $x_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ DESCRIBES A PARTICLE WITH $S_x = +\hbar/2$

ONE CAN SHOW THAT

$x_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $x_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ AND $y_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$ $y_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$

NOTE THAT

$(U_+)^\dagger U_+ = (U_-)^\dagger U_- = 1 \leftarrow$ NORMALIZATION

$(U_+)^\dagger U_- = (U_-)^\dagger U_+ = 0 \leftarrow$ ORTHOGONALITY

$x_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$ $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$x_- = \alpha z + \beta z$

$P(z_-) = |\beta|^2 = |1/\sqrt{2}|^2 = 1/2 = 0.5$

$\downarrow$

$-\hbar/2$

NOTATION

THE SPIN COMPONENT ALONG z

z_+ = IF I TRY TO MEASURE S_z, I WILL GET $+\hbar/2$

x_+ = IF I TRY TO MEASURE S_x, I WILL GET $+\hbar/2$

$v = \alpha z_+ + \beta z_-$ $\quad v = \alpha' x_+ + \beta' x_-$

PROBABILITY OF FINDING $+\hbar/2$ WHEN MEASURING ALONG X

THE TASK WOULD BE SIMPLE IF WE HAD V WRITTEN IN TERMS OF x_- AND x_+

$U_+^\dagger U_+ = 1$

$U_+^\dagger U_- = 0$

$$(x_+)^\dagger v = (x_+)^\dagger(\alpha' x_+ + \beta' x_-) = (x_+)^\dagger \alpha' x_+ + (x_+)^\dagger \beta' x_- = \alpha' \underbrace{(x_+)^\dagger x_+}_{1} + \beta' \underbrace{\cancel{(x_+^\dagger) x_-}}_{0} = \alpha'$$

$P = |\alpha'|^2$

$(x_+)^\dagger = (x_+)^\dagger(\alpha z_+ + \beta z_-) = \alpha (x_+)^\dagger z_+ + \beta (x_+)^\dagger z_-$

$x_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \to (x_+)^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \end{pmatrix}$

$z_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$z_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$x_+^\dagger v = \alpha \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\alpha}{\sqrt{2}} 1 + \frac{\beta}{\sqrt{2}}(-1) = \frac{1}{\sqrt{2}}(\alpha - \beta)$$

$$P_v(+\hbar/2) = |(x_+)^\dagger v|^2 = \left|\frac{1}{\sqrt{2}}(\alpha - \beta)\right|^2 = \frac{1}{2}\left(|\alpha|^2 + |\beta|^2 - 2\,\mathrm{Re}(\alpha\beta^*)\right)$$

AMPLITUDE VECTORS

AMPLITUDE VECTOR DESCRIBES THE PHYSICAL SITUATION OF THE SPIN 1/2 PARTICLE AS A SUPERPOSITION OF z_+ AND z_-

$\rightarrow S_v \quad P_+(-\hbar/2) = |(v_-)^\dagger v|^2$

STATE OF A QUANTUM SYSTEM

$v = \alpha z_+ + \beta z_- \qquad v = \alpha' x_+ + \beta' x_-$

$\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = \begin{pmatrix}(z_+)^\dagger v \\ (z_-)^\dagger v\end{pmatrix} \qquad \begin{pmatrix}\alpha' \\ \beta'\end{pmatrix} = \begin{pmatrix}(x_+)^\dagger v \\ (x_-)^\dagger v\end{pmatrix}$

- WE STRESS THAT W/ V WE ARE INDICATING THE SAME PHYSICAL SITUATION WHILE THE AMPLITUDE VECTOR DEPENDS ON THE BASIS WE ARE CONSIDERING
- WE USE TERM **STATE** TO MEAN A PHYSICAL SITUATION FOR A GIVEN QUANTUM SYSTEM, REGARDLESS OF ITS REPRESENTATION

IF I CHANGE MY REFERENCE FRAME THE REPRESENTATION CHANGE BUT THE STATE DO NOT

PAUL DIRAC INTRODUCED A CLEVER NOTATION TO INDICATE THE STATE OF A QUANTUM SYSTEM AND AVOID ANY CONFUSION

$| \ \rangle$ CALLED THE KET $\longrightarrow$ INDICATE THE STATE

A SITUATION IN WHICH A SPIN -1/2 PARTICLE HAS $S_z = +\hbar/2$ MIGHT BE WRITTEN

$|z_+\rangle$ OR $|\text{SPIN UP}\rangle$ OR $|\uparrow\rangle$

FOR EXAMPLE

$x_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1 \\ 1\end{pmatrix} \qquad |x_+\rangle = \frac{1}{\sqrt{2}}|z_+\rangle + \frac{1}{\sqrt{2}}|z_-\rangle$

$\downarrow$

REPRESENTATION IN THE z_+ AND z_- BASIS

$v \rightarrow |v\rangle$ THIS CALLED KET $\qquad u^\dagger v \longrightarrow \langle u|v\rangle$ BRACKET

$u^\dagger \rightarrow \langle u|$ BRA

$x_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1 \\ 1\end{pmatrix} = \begin{pmatrix}\langle z_+|x_+\rangle \\ \langle z_-|x_+\rangle\end{pmatrix} \qquad |x_+\rangle = \frac{1}{\sqrt{2}}|z_+\rangle + \frac{1}{\sqrt{2}}|z_-\rangle$

RULE TO DETERMINE THE AMPLITUDE VECTOR V FOR A GIVEN STATE AND A GIVEN BASIS (v_+, v_-)

$|v\rangle = \alpha_+|v_+\rangle + \alpha_-|v_-\rangle \qquad v = \begin{pmatrix}\alpha_+ \\ \alpha_-\end{pmatrix} = \begin{pmatrix}\langle v_+|v\rangle \\ \langle v_-|v\rangle\end{pmatrix}$

$|v\rangle = \alpha|v_+\rangle + \beta|v_-\rangle$

CAN BE MOVED OUT (ARE NUMBERS)

CALCULATE $\langle v_+|v\rangle = \langle v_+|\alpha|v_+\rangle + \langle v_+|\beta|v_-\rangle$

$\langle v_+|v\rangle = (v_+)^\dagger v$

$= \alpha\langle v_+|v_+\rangle + \beta\langle v_+|v_-\rangle$

$= \alpha\underbrace{(v_+)^\dagger v_+}_{1} + \beta\underbrace{(v_+)^\dagger v_-}_{0}$

$= \boxed{\alpha}$